Stability and Growth in Incompressible Fluid Equations

Lecture Notes from the 2026 Yunnan University Summer School

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Abstract

These notes reorganize four lectures on stability, long-time dynamics, and small-scale formation in incompressible fluid equations. The central theme is that a conserved or monotone quantity first confines a solution near a coherent structure, while transport, hyperbolicity, or gravity then forces growth in a stronger norm. The examples include the two-dimensional Euler equation on the plane, torus, and a moving domain; the three-dimensional axisymmetric Euler equation without swirl; the incompressible porous media equation; and viscous and inviscid Boussinesq systems. The statements and proof sketches follow the June 2026 lectures; open questions are recorded in that historical context.

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1 The common framework

1.1 Transport and diagnostic velocity laws

Many incompressible models can be written as

$$\partial_t q + u \cdot \nabla q = F, \qquad \nabla \cdot u = 0, \tag{1.1}$$

where the scalar or vector q is transported by a velocity u. The models differ through the forcing F and the diagnostic law that recovers u from q. Incompressibility makes the Lagrangian flow map Φ_t measure preserving. Thus, when $F = 0$, every distributional invariant $\int f(q(x,t))\,\mathrm{d}x$ is conserved. This simple observation underlies both rearrangement stability and the growth mechanisms discussed below.

The recurring question is not merely whether a smooth solution exists for all time. It is whether a globally regular solution can develop progressively smaller spatial scales, as measured by unbounded Sobolev or Lipschitz norms. For a transported scalar, growth of ∇q measures deformation of level sets. For axisymmetric Euler flow, growth of vorticity records radial stretching of vortex rings. In the gravity-driven systems, potential energy provides a monotone quantity that forces mixing or instability.

1.2 Two-dimensional Euler equation

On $\mathbb{R}^2$, the vorticity form of the Euler equation is

$$\partial_t \omega + u \cdot \nabla \omega = 0, \qquad u = \nabla^{\perp}(-\Delta)^{-1}\omega = \frac{1}{2\pi}\int_{\mathbb{R}^2} \frac{(x-y)^{\perp}}{|x-y|^2}\omega(y)\,\mathrm{d}y. \tag{1.2}$$

The flow conserves the distribution function of ω, kinetic energy

$$E[\omega] = \frac{1}{2}\int_{\mathbb{R}^2} |u|^2\,\mathrm{d}x, \tag{1.3}$$

and, under suitable decay, the first moment and moment of inertia. For data odd in x_2, it is convenient to use the vertical impulse

$$\mu[\omega] = \int_{\mathbb{R}^2_+} x_2\omega(x)\,\mathrm{d}x, \qquad \mathbb{R}^2_+ = \{x_2 > 0\}. \tag{1.4}$$

Classical Yudovich theory gives global uniqueness for bounded vorticity (Yudovich, 1963). The log-Lipschitz character of the velocity implies a double-exponential upper bound for $\|\nabla\omega(t)\|_{L^\infty}$ when the initial gradient is bounded. Determining which lower growth rates can actually occur is substantially harder.

> **Definition 1.1** (Steady and relative equilibria)**.** A vorticity ω_s is a steady state if the velocity it generates satisfies $u_s \cdot \nabla\omega_s = 0$. A relative equilibrium is stationary only after a translation or rotation. A traveling dipole has the form $\omega(x,t) = \omega_D(x - cte_1)$.

Every radially symmetric vorticity is steady. If $\omega_s \geq 0$ is radially decreasing, its radially decreasing rearrangement minimizes the moment of inertia and maximizes the interaction energy within its rearrangement class. This yields nonlinear L^1 stability for nonnegative perturbations (Wan and Pulvirenti, 1985). The sign-changing problem is subtler and remains a useful warning: rearrangement control is strongest when positivity is preserved.

2 Lamb dipoles and vortex quadrupoles

2.1 The Lamb dipole as an energy maximizer

A vortex dipole consists of two counter-rotating vortices and travels at a constant speed. The Lamb–Chaplygin dipole ω_L is compactly supported, odd in x_2, and nonnegative on $\mathbb{R}^2_+$. Its variational characterization is more important here than its Bessel-function formula.

Theorem 2.1 (Variational stability of the Lamb dipole)**.** *Within the class of vorticities that are odd in x_2 and nonnegative on $\mathbb{R}^2_+$, the Lamb dipole uniquely maximizes kinetic energy subject to fixed impulse and enstrophy constraints, up to horizontal translation. Quantitative coercivity of the energy deficit controls the distance to the orbit $\{\omega_L(\cdot - ae_1) : a \in \mathbb{R}\}$ in a weighted norm.*

This result, proved in a precise form by Abe and Choi (2022), converts the conservation laws into orbital stability. Schematically, with

$$\|f\|_X := \|f\|_{L^2} + \int_{\mathbb{R}^2} |x_2 f(x)| \, \mathrm{d}x, \tag{2.1}$$

one obtains an increasing function F with $F(0) = 0$ such that

$$E[\omega_L] - E[\omega] \gtrsim F\left(\inf_{a\in\mathbb{R}} \|\omega - \omega_L(\cdot - ae_1)\|_X\right). \tag{2.2}$$

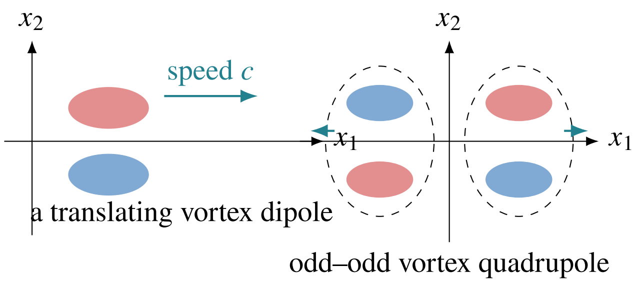


Figure 1: A traveling dipole and the odd–odd quadrupole obtained by reflection. The colors denote opposite signs of vorticity, not different fluids.

2.2 Odd–odd symmetry and the interaction energy

Let $Q = \{x_1 > 0, x_2 > 0\}$ and assume ω_0 is odd in both coordinate directions and nonnegative in Q. Write $\rho = \omega \mathbf{1}_Q$. Reflection produces four vortices with alternating signs. When the two dipoles separate horizontally, the right component ω^r remains close to a Lamb dipole.

Let $N(z) = -(2\pi)^{-1} \log |z|$ be the Newtonian kernel. The energy splits into a dipole part and a cross-interaction part:

$$E[\omega] = 2 \iint \omega^r(x)\omega^r(y)N(x-y)\,\mathrm{d}x\,\mathrm{d}y \tag{2.3}$$

$$- 2 \iint \omega^r(x)\omega^\ell(y)N(x-y)\,\mathrm{d}x\,\mathrm{d}y =: E_{\mathrm{dip}}(t) + E_{\mathrm{int}}(t). \tag{2.4}$$

Odd–odd symmetry gives $E_{\mathrm{int}}(t) > 0$. If the initial separation is $d \gg 1$, then $E_{\mathrm{int}}(0) = O(d^{-2})$. Energy conservation therefore prevents the dipole energy from falling by more than $O(d^{-2})$. In addition, the right impulse

$$\mu[\omega^r] = \int_{\mathbb{R}^2} x_2 \omega^r(x)\,\mathrm{d}x \tag{2.5}$$

is nonincreasing in this symmetry class. The variational stability theorem then keeps each departing dipole close to a translate of ω_L.

2.3 Concentrated quadrupoles

The point-vortex orbit in the first quadrant starting from $p = (p_1, p_2)$ is

$$O_{\mathrm{pv}}(p) = \left\{ x \in Q : \frac{2(x_1x_2)^2}{x_1^2 + x_2^2} = \frac{2(p_1p_2)^2}{p_1^2 + p_2^2} \right\}. \tag{2.6}$$

Finite-time desingularization says that a sufficiently concentrated smooth vortex follows this orbit on every fixed time interval. The central issue is to make the conclusion global in time.

Theorem 2.2 (Global concentrated-vortex dynamics)**.** *Let ω_0 be odd–odd and set $\rho(t) = \omega(t)\mathbf{1}_Q$, with $\int_Q \rho = 1$. Given $\varepsilon > 0$, $A \geq 1$, and $p \in Q$, there exists $\delta_0 = \delta_0(\varepsilon, A, p) > 0$ such that, for $0 < \delta < \delta_0$, the conditions*

$$\operatorname{supp} \rho_0 \subset B(p, \delta), \qquad 0 \leq \rho_0 \leq A\delta^{-2}, \qquad \left\|\rho_0 - \rho_0^*(\cdot - p)\right\|_{L^1} \leq \delta \tag{2.7}$$

imply, for all $t \in \mathbb{R}$,

$$\inf_{a\in\mathbb{R}^2} \left\|\rho(t) - \rho_0^*(\cdot - a)\right\|_{L^1} \leq \varepsilon, \tag{2.8}$$

$$\operatorname{dist}\big(X(t), O_{\mathrm{pv}}(p)\big) \leq \varepsilon, \qquad X(t) = \int_Q x\rho(x,t)\,\mathrm{d}x. \tag{2.9}$$

Here ρ_0^* is the radially decreasing rearrangement of ρ_0.

The result is due to Choi et al. (2024). Its conclusion is both a global desingularization theorem and an orbital stability statement for a general concentrated, nearly radially decreasing profile.

2.4 Why a direct rearrangement argument fails

For a profile at height $h_0 \gg 1$, the reflected dipole kernel is

$$\log \frac{|x - \bar{y}|}{|x - y|}, \qquad \bar{y} = (y_1, -y_2). \tag{2.10}$$

Splitting this into a self-energy and an image interaction is tempting. The self-energy is maximized by symmetric decreasing rearrangement, with a sharp deficit estimate (Yan and Yao, 2022); the image interaction, however, is unbounded above under a fixed impulse. It must be treated together with the self-energy.

The useful pointwise bound has the schematic form

$$\log \frac{|x - \bar{y}|}{|x - y|} \leq \varphi(|x - y|) + g^{h_0}(2x_2) + O(h_0^{-1}), \tag{2.11}$$

where φ is radial and decreasing, while g^{h_0} is increasing and concave. Jensen's inequality and impulse monotonicity control the second term; the sharp rearrangement inequality controls the first. Combining the upper bound with the lower bound supplied by energy conservation yields

$$\inf_{a\in\mathbb{R}^2} \left\|\rho(t) - \rho_0^*(\cdot - a)\right\|_{L^1}^2 \lesssim \frac{h_0}{\ell_0} + h_0^{-1}, \tag{2.12}$$

when the vortex begins near (ℓ_0, h_0) with $\ell_0 \gg h_0 \gg 1$.

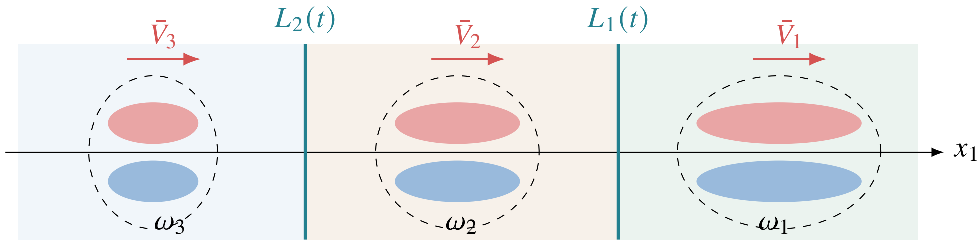
To pass from a far-away reference point to an arbitrary $p \in Q$, one chooses a small stable packet near a distant point q, uses finite-time desingularization to map a packet near p into that stable neighborhood, and then runs the positive-time stability argument. Time reversibility supplies the negative-time half. Gluing constructions provide complementary special solutions with prescribed asymptotics (Dávila et al., 2023).

3 Several Lamb dipoles with ordered speeds

Consider N Lamb dipoles traveling to the right with distinct ordered speeds

$$\bar{V}_1 > \bar{V}_2 > \cdots > \bar{V}_N. \tag{3.1}$$

The rightmost dipole is the fastest, so the configuration separates rather than collides. The stability theorem of Abe et al. (2025) shows that a sufficiently separated, sufficiently small perturbation remains close to this multi-dipole orbit for all positive time.



moving borders separate components with ordered speeds

Figure 2: The solution is decomposed by moving borders. The speed ordering prevents one coherent component from overtaking the component ahead.

Let $L_i(t)$ start at the midpoint between neighboring dipoles and move with

$$L_i'(t) = \frac{1}{2}(\bar{V}_i + \bar{V}_{i+1}). \tag{3.2}$$

Define ω_i to be the portion of the vorticity in the corresponding moving cell. For each component track the enstrophy, impulse, and kinetic energy

$$K_i(t) = \int \omega_i^2 \,\mathrm{d}x, \qquad \mu_i(t) = \int x_2\omega_i \,\mathrm{d}x, \qquad E_i(t) = \frac{1}{2}\int |u_i|^2 \,\mathrm{d}x. \tag{3.3}$$

These quantities are not exactly conserved because vorticity crosses the borders and the Biot–Savart interaction is nonlocal. For $N = 2$, a typical bootstrap regime is

$$K_1(t) \le K_1(0) + \delta, \quad \mu_1(t) \le \mu_1(0) + \delta, \quad E_i(t) \ge E_i(0) - \delta \quad (i = 1, 2). \tag{3.4}$$

The delicate term is the positive contribution of the left dipole to the right impulse. A pointwise small bound on $\mu_1'(t)$ is insufficient: its time integral must be small. The proof follows every particle that crosses a border, records its subsequent gain or loss, and uses the speed gap to show that the total Lagrangian contribution is summable. Once (3.4) improves itself, the Lamb energy coercivity applies component by component.

4 From orbital stability to gradient growth

4.1 Landmarks and the hyperbolic mechanism

Growth examples for the two-dimensional Euler equation include a hyperbolic steady flow on the torus (Bahouri and Chemin, 1994), superlinear growth on $\mathbb{T}^2$ (Denisov, 2009), double-exponential growth in a disk (Kiselev and Šverák, 2014), exponential growth on $\mathbb{T}^2$ for $C^{1,\alpha}$ data (Zlatoš, 2015), and maximal double-exponential growth on the half-plane (Zlatoš, 2025). Before the work described below, no compactly supported example on $\mathbb{R}^2$ was known to grow faster than linearly.

The common geometry is a saddle point. Suppose a steady or relative equilibrium is orbitally stable and its velocity has a hyperbolic stagnation point. A small perturbation can carry a level-set filament through a rectangle around the saddle. Uniform inward flux through two sides and outward flux through the other two sides forces the filament width $h(t)$ to shrink. Since the scalar jump is fixed,

$$\|\nabla\omega(t)\|_{L^\infty} \gtrsim h(t)^{-1}. \tag{4.1}$$

If a positive amount of flux crosses for all time, an integral argument gives $h(t) \lesssim (1+t)^{-1}$ along a sequence and hence superlinear growth after a slightly sharper construction.

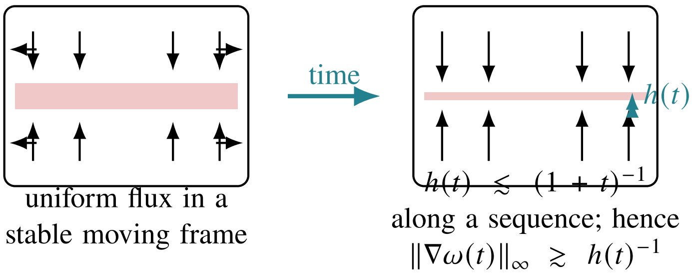


Figure 3: Orbital stability preserves the sign of the boundary flux in a suitable frame; incompressibility then forces transverse thinning.

4.2 A moving frame on the plane

For a perturbed Lamb dipole, the obvious choices of center fail. The center of mass is too sensitive to a long, light tail; an L^2 minimizing translation need not depend regularly on time. Instead choose a smooth increasing function g satisfying

$$g(s) = s \quad (|s| \le 2), \qquad g(s) = 3 \quad (s \ge 3), \qquad g(s) = -3 \quad (s \le -3), \tag{4.2}$$

and define $p(t)$ implicitly by

$$\int_{\mathbb{R}^2_+} \omega(t, x + te_1) g(x_1 - p(t))\, \mathrm{d}x = 0. \tag{4.3}$$

The truncation makes $p(t)$ insensitive to distant tails. Orbital stability shows that $p(t)$ remains close to any admissible orbital translation, while differentiating (4.3) and using the Euler equation gives $|p'(t)| \ll 1$. Thus the hyperbolic rectangle moves almost with the Lamb dipole and retains a uniform flux pattern.

Theorem 4.1 (Superlinear growth without boundary)**.** *There exist smooth, compactly supported initial vorticities on $\mathbb{R}^2$, close to a Lamb dipole and odd in x_2, for which $\|\nabla\omega(t)\|_{L^\infty}$ grows superlin-*

> *early along an unbounded time sequence. On $\mathbb{T}^2$, an orbitally stable steady state up to translation with a saddle point generates an open set of smooth initial data with superlinear gradient growth.*

The whole-plane and torus statements are proved in Jeong et al. (2025). On the torus, a regular translation vector can be defined by Fourier phases. If $k_1 = (1,0)$ and $k_2 = (0,1)$ and the corresponding Fourier coefficients of the steady state are nonzero, set

$$p_j(t) = \arg \widehat{\omega}_*(k_j) - \arg \widehat{\omega}(t, k_j), \qquad j = 1, 2. \tag{4.4}$$

Higher modes replace a vanishing first coefficient. Differentiation again shows $|p'(t)| \ll 1$.

> **Question 4.2** (Gradient growth without a boundary)**.** As posed in the June 2026 lectures: can one obtain an open set of superlinearly growing data on $\mathbb{R}^2$ without a symmetry assumption? On $\mathbb{R}^2$ or $\mathbb{T}^2$, can one construct growth faster than superlinear, even $t^{1+\varepsilon}$?

5 Two-dimensional Euler flow with a free boundary

5.1 The fixed-domain benchmark

Let D be a fixed smooth bounded domain. The Lagrangian flow satisfies

$$\frac{\mathrm{d}}{\mathrm{d}t}\Phi_t(x) = u(\Phi_t(x), t), \qquad \Phi_0(x) = x. \tag{5.1}$$

Because vorticity is constant along trajectories,

$$\|\nabla\omega(t)\|_{L^\infty} \le \|\nabla\omega_0\|_{L^\infty} \exp\left(\int_0^t \|\nabla u(s)\|_{L^\infty}\, \mathrm{d}s\right). \tag{5.2}$$

Kato's logarithmic estimate

$$\|\nabla u\|_{L^\infty} \le C\,\|\omega\|_{L^\infty}\left(1 + \log_+ \|\nabla\omega\|_{L^\infty}\right) \tag{5.3}$$

and Gronwall's inequality give a double-exponential upper bound. The disk construction of Kiselev and Šverák (2014) realizes this scale. Near a boundary hyperbolic point one has, schematically,

$$u_1(x) = -\frac{4}{\pi}x_1\Omega(x) + x_1B_1(x), \qquad u_2(x) = \frac{4}{\pi}x_2\Omega(x) + x_2B_2(x), \tag{5.4}$$

where

$$\Omega(x) = \int_{Q(x)} \frac{y_1y_2}{|y|^4}\omega(y)\,\mathrm{d}y, \qquad |B_1| + |B_2| \le C. \tag{5.5}$$

A trapezoid on which $\omega = 1$ makes Ω grow like $\log(b(t)/a(t))$, and the ratio $b(t)/a(t)$ becomes double exponential.

5.2 Moving-domain formulation

Let D_t be the fluid region, with fixed bottom Γ_b and free surface Γ_t. With surface tension coefficient $\sigma > 0$, the equations are

$$\begin{cases} \partial_t u + (u\cdot\nabla)u = -\nabla p, & x \in D_t, \\ \nabla\cdot u = 0, & x \in D_t, \\ u\cdot n = 0, & x \in \Gamma_b, \\ V_{\Gamma_t} = u\cdot n, \quad p = \sigma\mathcal{H}, & x \in \Gamma_t, \end{cases} \tag{5.6}$$

where $\mathcal{H}$ is the curvature and the free surface moves with normal velocity V_{Γ_t}. Vorticity is still transported, but there is no Biot–Savart law determined by ω alone: adding $\nabla^\perp\phi$ for a harmonic ϕ with compatible bottom data preserves vorticity.

Theorem 5.1 (Double-exponential growth with a free boundary)**.** *Set $\sigma = 1$ and let $D_0 = \mathbb{T}\times[0,2]$, with the flat top as free surface. There is a smooth velocity $v_0 \in C^\infty(D_0)$ and universal constants $\varepsilon_0, c_1, c_2 > 0$ such that, for every $0 < \varepsilon < \varepsilon_0$, the solution with $u_0 = \varepsilon v_0$ satisfies*

$$\|\nabla\omega(t)\|_{L^\infty(D_t)} \ge \varepsilon\exp(c_1\exp(c_2\varepsilon t)), \qquad 0 \le t < T, \tag{5.7}$$

where T is the lifespan of the smooth solution.

This theorem is due to Hu et al. (2024). Equivalently, data of size $O(\varepsilon)$ can reach order-one $W^{2,\infty}$ size on the scale $t \sim \varepsilon^{-1}\log\log\varepsilon^{-1}$, unless a singularity occurs sooner.

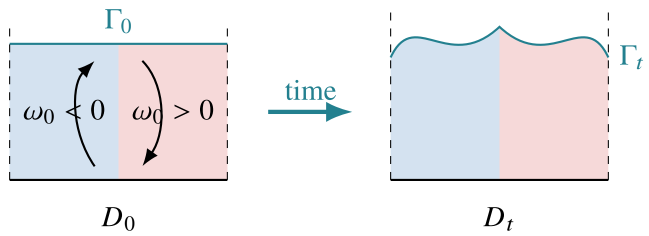


Figure 4: The initial flat strip and a possible later free surface. Odd symmetry in x_1 is preserved, while surface tension controls the total excess length rather than the pointwise graph.

5.3 Proof architecture

The first input is energy conservation:

$$K(t) + \sigma L(t) = K(0) + \sigma L(0), \tag{5.8}$$

where $K(t) = \frac{1}{2}\int_{D_t} |u|^2$ and $L(t)$ is the free-surface length. The flat surface minimizes length. Since $K(0) = O(\varepsilon^2)$, (5.8) gives $K(t) \le K(0)$ and $L(t) \le L(0) + O(\varepsilon^2)$. Symmetry and this length deficit keep the surface above a fixed interior strip containing the hyperbolic point.

In that fixed strip $Q = \mathbb{T} \times [0, 1]$, construct $U = \nabla^\perp \Psi$ by

$$\Delta\Psi = \omega \mathbf{1}_Q, qquad \Psi\big|_{\partial Q} = 0. \tag{5.9}$$

The symmetry of ω makes U hyperbolic near the origin; for example, $U_1(x, 0) \le -c\varepsilon x_1$ for $0 < x_1 \ll 1$. Set $e = u - U$ in Q. Hodge orthogonality and the energy bound give $\|e\|_{L^2(Q)} \lesssim \varepsilon$. Moreover, e is harmonic after reflection across the bottom, so interior elliptic estimates yield

$$\|\nabla e\|_{L^\infty(B_{1/2})} \lesssim \|e\|_{L^2(Q)} \lesssim \varepsilon. \tag{5.10}$$

Thus the true velocity retains the hyperbolic structure. The Kiselev–Šverák trajectory argument then gives the lower bound in theorem 5.1.

Adding gravity changes the conserved energy to $K + P + \sigma L$, and the flat reference domain also minimizes the gravitational potential energy. Removing surface tension, moving to infinite depth, or forcing growth directly near the free surface are harder questions recorded in the lectures.

6 Axisymmetric Euler flow without swirl

6.1 Reduction and the Childress scenario

In cylindrical coordinates (r, θ, z), an axisymmetric velocity without swirl and its vorticity are

$$u = u^r(r, z)e_r + u^z(r, z)e_z, \qquad \omega = \omega^\theta(r, z)e_\theta, \qquad \omega^\theta = -\partial_z u^r + \partial_r u^z. \tag{6.1}$$

The relative vorticity

$$\xi = \frac{\omega^\theta}{r} \tag{6.2}$$

obeys a pure transport equation,

$$\partial_t \xi + u^r \partial_r \xi + u^z \partial_z \xi = 0. \tag{6.3}$$

Thus growth of $\omega^\theta = r\xi$ can only occur when vorticity is carried to large radius. This class is globally well posed under standard integrability and regularity assumptions on ω_0^θ and ω_0^θ/r.

The model configuration has ω_0^θ odd in z and nonnegative in

$$\Pi_+ = \{(r,z) : r > 0,\ z > 0\}. \tag{6.4}$$

It represents two identical, oppositely signed vortex rings. Experiments and numerics suggest that they approach the symmetry plane and spread radially. Childress (2008) conjectured the rate $\|\omega(t)\|_{L^\infty} \sim t^{4/3}$.

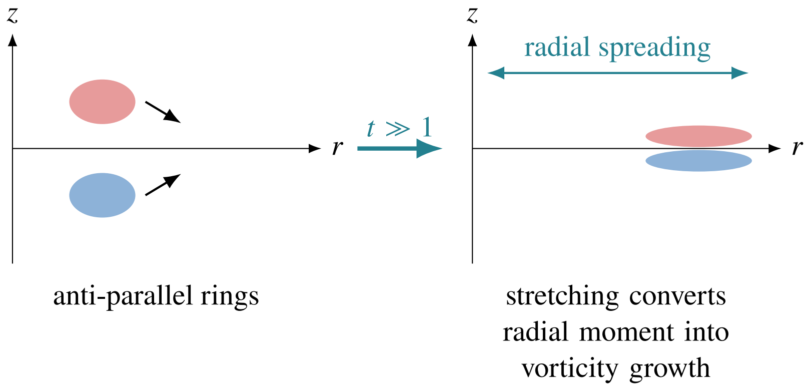


Figure 5: In a meridional half-plane, two anti-parallel vortex rings approach $z = 0$ and spread toward larger r. Conservation of ω^θ/r converts radial displacement into vorticity amplification.

6.2 Monotone moments

Define radial and vertical moments on Π_+ by

$$P_2(t) = \iint_{\Pi_+} r^2\omega^\theta(r,z,t)\,\mathrm{d}r\,\mathrm{d}z, \qquad Z(t) = \iint_{\Pi_+} z\omega^\theta(r,z,t)\,\mathrm{d}r\,\mathrm{d}z. \tag{6.5}$$

In the Childress sign class, $P_2'(t) > 0$ and $Z'(t) < 0$. Earlier lower bounds follow by expressing these derivatives as double integrals and proving a sign for the kernels; see, for example, Gustafson et al. (2026). The lectures emphasized two velocity identities that are especially efficient:

$$P_2'(t) = 2\iint_{\Pi_+} ru^r\omega^\theta\,\mathrm{d}r\,\mathrm{d}z = \int_0^\infty r\,u^r(r,0,t)^2\,\mathrm{d}r > 0, \tag{6.6}$$

$$-Z'(t) = \iint_{\Pi_+} u^z\omega^\theta\,\mathrm{d}r\,\mathrm{d}z = \frac{1}{2}\int_0^\infty u^z(0,z,t)^2\,\mathrm{d}z + \iint_{\Pi_+} \frac{(u^r)^2}{r}\,\mathrm{d}r\,\mathrm{d}z > 0. \tag{6.7}$$

Theorem 6.1 (Moment and norm growth)**.** *Assume $u_0 \in L^2(\mathbb{R}^3)$, $\omega_0^\theta, \omega_0^\theta/r \in L^1 \cap L^\infty(\mathbb{R}^3)$, $P_2(0), Z(0) < \infty$, and that ω_0^θ is odd in z and positive in Π_+. Then*

$$\frac{t}{\log(2+t)} \lesssim P_2(t) \lesssim (1+t)^2. \tag{6.8}$$

For every $1 \le p \le \infty$,

$$\limsup_{t\to\infty} \frac{\|\omega^\theta(t)\|_{L^p(\mathbb{R}^3)}}{t^{1/4}} \ge c(u_0,\omega_0) > 0. \tag{6.9}$$

Smooth compactly supported initial vorticity is allowed.

The result is from Egamberganov and Yao (2025). The upper moment bound follows from

$$P_2'(t) \le 2 \iint r \left|u^r \omega^\theta\right| \, \mathrm{d}r \, \mathrm{d}z \le C P_2(t)^{1/2}, \tag{6.10}$$

where energy and the transported L^∞ norm of ω^θ/r control the remaining factor. The optimal velocity interpolation behind the $t^{4/3}$ vorticity upper bound is developed in Lim and Jeong (2024).

For the lower bound, the axisymmetric Biot–Savart formula and monotonicity of Z give

$$\int_0^\infty u^r(r,0,t)\, \mathrm{d}r = \iint_{\Pi_+} \left(1 - \frac{z}{\sqrt{r^2+z^2}}\right) \omega^\theta \, \mathrm{d}r \, \mathrm{d}z \ge \gamma(\omega_0) > 0. \tag{6.11}$$

Choosing $a(t) \sim (1+t)^{-1/3}$ and $b(t) \sim 1+t$, Cauchy–Schwarz gives

$$P_2'(t) \gtrsim \frac{\left(\int_{a(t)}^{b(t)} u^r(r,0,t)\, \mathrm{d}r\right)^2}{\log(b(t)/a(t))} \gtrsim \frac{1}{\log(2+t)}. \tag{6.12}$$

For norm growth, (6.7) yields a local-mass inequality: if $\iint_{r<R} \omega^\theta \, \mathrm{d}r \, \mathrm{d}z > m$, then

$$-Z'(t) \ge c_0(u_0, \omega_0) m^4 R^{-4}. \tag{6.13}$$

Because $-Z'$ is integrable in time, taking $R \sim t^{1/4}$ shows that at some $t_R \in [0,t]$ most vorticity lies beyond R. Conservation of total relative vorticity then forces the L^p lower bound. A byproduct is the time-integrated escape statement

$$\int_0^\infty m_R(t)^4 \, \mathrm{d}t \le C Z(0) R^4, \qquad m_R(t) = \iint_{r<R} \omega^\theta(r,z,t)\, \mathrm{d}r \, \mathrm{d}z. \tag{6.14}$$

Question 6.2 (Axisymmetric growth)**.** Can the lower bound for P_2 be improved toward the conjectural t^2 scale? Can one obtain a sharper lower bound for the maximal support radius than $R(t) \gtrsim P_2(t)^{1/2}$? Can the exponent $1/4$ in theorem 6.1 be improved, or can the limsup be replaced by a limit?

7 The incompressible porous media equation

7.1 Darcy transport and comparison with Euler

Let ρ be density and u the Darcy velocity. After normalizing permeability, viscosity, and gravity, the incompressible porous media (IPM) system is

$$\partial_t \rho + u \cdot \nabla \rho = 0, \qquad \nabla \cdot u = 0, \qquad u = -\nabla p - \rho e_2. \tag{7.1}$$

Taking the scalar curl of Darcy's law gives $\operatorname{curl} u = -\partial_{x_1} \rho$. Hence

$$u = \partial_{x_1} \nabla^\perp (-\Delta)^{-1} \rho. \tag{7.2}$$

The horizontal derivative appears because pressure removes the gradient part of the vertical buoyancy force. The diagnostic laws for three transported scalars are usefully compared as follows:

Model	Velocity law	Fourier order
2D Euler	$u = \nabla^\perp(-\Delta)^{-1}\rho$	-1
SQG	$u = \nabla^\perp(-\Delta)^{-1/2}\rho$	0
2D IPM	$u = \partial_{x_1}\nabla^\perp(-\Delta)^{-1}\rho$	0

Local well-posedness and continuation criteria were established by Córdoba et al. (2007); global regularity for arbitrary smooth data remains open. Unlike 2D Euler, the odd–odd hyperbolic symmetry is not preserved. Horizontal stratifications $\rho_s = g(x_2)$ are steady and become the natural reference states.

7.2 Potential energy forces small scales

Assume first that $\Omega = \mathbb{R}^2$, ρ_0 is odd in x_2, nonnegative in the upper half-plane, smooth, and compactly supported. Define potential energy

$$\mathcal{E}(t) = \int_{\mathbb{R}^2} \rho(x,t)x_2\,\mathrm{d}x. \tag{7.3}$$

Using transport, incompressibility, and (7.2),

$$\mathcal{E}'(t) = \int_{\mathbb{R}^2} \rho u_2\,\mathrm{d}x = \int_{\mathbb{R}^2} \rho\,\partial_{x_1}^2(-\Delta)^{-1}\rho\,\mathrm{d}x \tag{7.4}$$

$$= -\left\|\partial_{x_1}\rho(t)\right\|_{\dot H^{-1}}^2 =: -\delta(t). \tag{7.5}$$

Since $\mathcal{E}(t) \geq 0$, one obtains

$$\int_0^\infty \delta(t)\,\mathrm{d}t \leq \mathcal{E}(0). \tag{7.6}$$

There is no physical diffusion: the apparent dissipation is a monotonicity identity for potential energy.

Fourier splitting supplies the complementary estimate

$$\|\rho(t)\|_{\dot H^s} \gtrsim \delta(t)^{-s/4}, \qquad s > 0. \tag{7.7}$$

The L^2 conservation and the L^1 bound on $\widehat{\rho}$ prevent Fourier mass from concentrating entirely in the narrow cone where $\xi_1^2/|\xi|^2$ is small. Combining (7.6) and (7.7) gives the theorem.

Theorem 7.1 (IPM small-scale formation on $\mathbb{R}^2$)**.** *Under the preceding symmetry and sign assumptions, if the IPM solution stays smooth for all time, then for every $s > 0$,*

$$\int_0^\infty \|\rho(t)\|_{\dot H^s(\mathbb{R}^2)}^{-4/s}\,\mathrm{d}t \leq C(s,\rho_0) < \infty. \tag{7.8}$$

Consequently,

$$\limsup_{t\to\infty} t^{-s/4}\,\|\rho(t)\|_{\dot H^s(\mathbb{R}^2)} = \infty. \tag{7.9}$$

On $\mathbb{T}^2$, impose that ρ_0 is odd in x_2, even in x_1, vanishes on $x_1 = 0$, and is nonnegative on $[0,\pi]^2$. A periodic version of the energy identity holds. The zero set on the x_2-axis forces a large horizontal derivative when the level sets become nearly horizontal. For $s > -1/2$,

$$\int_0^\infty \left\|\partial_{x_1}\rho(t)\right\|_{\dot H^s(\mathbb{T}^2)}^{-2/(2s+1)}\,\mathrm{d}t < \infty. \tag{7.10}$$

In particular, for $s > 1/2$,

$$\limsup_{t\to\infty} t^{-(s-1/2)}\,\|\rho(t)\|_{\dot H^s(\mathbb{T}^2)} = \infty. \tag{7.11}$$

The same construction descends to a strip with no-flow top and bottom. These results are due to Kiselev and Yao (2023).

7.3 Stability and instability of stratified states

Write $\rho = \rho_s + \eta$ with $\rho_s = g(x_2)$. Then

$$\partial_t \eta + u \cdot \nabla \eta = -g'(x_2) u_2, \qquad u = \partial_{x_1} \nabla^{\perp} (-\Delta)^{-1} \eta. \tag{7.12}$$

The linearized operator is $-g'(x_2)(-\Delta)^{-1}\partial_{x_1}^2$. The stable profile $\rho_s = -x_2$ is asymptotically stable in sufficiently strong Sobolev spaces, but the same state is unstable in weaker topologies. Thus "stability" is not meaningful until the norm is specified.

Theorem 7.2 (Nonlinear instability of stratifications)**.** *On $\mathbb{T}^2$, let $\rho_s = g(x_2) \in C^\infty$ be odd. For every $k > 0$ and $\varepsilon > 0$ there exists smooth ρ_0 with $\|\rho_0 - \rho_s\|_{H^k} \le \varepsilon$ such that, if the solution remains smooth,*

$$\limsup_{t\to\infty} t^{-s/2} \|\rho(t) - \rho_s\|_{\dot{H}^{s+1}} = \infty \qquad (s > 0). \tag{7.13}$$

On a strip $S = \mathbb{T} \times [-\pi, \pi]$, for every stationary ρ_s and $\varepsilon, \gamma > 0$, one can choose smooth data with $\|\rho_0 - \rho_s\|_{H^{2-\gamma}(S)} \le \varepsilon$ and the same growth conclusion.

For an odd periodic g, some height h_0 has $g'(h_0) > 0$. A small circular transport near $(0, h_0)$ lowers the potential energy by a fixed amount while remaining arbitrarily small in every prescribed H^k norm. The energy gap prevents the solution from returning too close to the stratified orbit. In a strip, even the linearly stable profile $-x_2$ can be perturbed by a small "bubble" that creates a closed level set. Transport preserves this closed component, again obstructing convergence in a weak Sobolev topology.

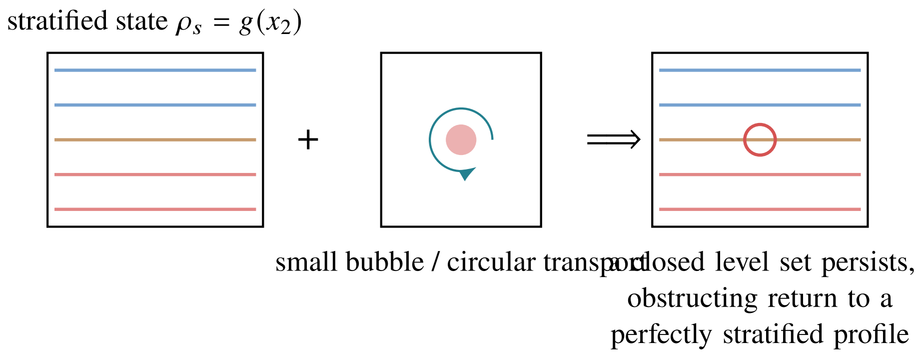


Figure 6: Two perturbations used in the IPM instability argument. A local circular rearrangement changes potential energy; a small bubble creates a topological obstruction that survives transport.

8 Viscous and inviscid Boussinesq equations

8.1 Equations and energy

Without density diffusion, the two-dimensional Boussinesq system is

$$\begin{cases} \partial_t \rho + u \cdot \nabla \rho = 0, \\ \partial_t u + u \cdot \nabla u = -\nabla p - \rho e_2 + \nu \Delta u, \\ \nabla \cdot u = 0. \end{cases} \tag{8.1}$$

The case $\nu > 0$ is viscous; $\nu = 0$ is inviscid. Global regularity for the viscous system with smooth data is classical; see, for example, Hou and Li (2005). The density is not diffused, so its Sobolev norms can still grow without bound.

On the torus, define

$$E_P(t) = \int_{\mathbb{T}^2} \rho x_2 \,\mathrm{d}x, \qquad E_K(t) = \frac{1}{2}\int_{\mathbb{T}^2} |u|^2 \,\mathrm{d}x. \tag{8.2}$$

In the symmetric setting used below, the coordinate representative of x_2 is compatible with the sign conditions. The total energy satisfies

$$\frac{\mathrm{d}}{\mathrm{d}t}(E_P + E_K) = -\nu \|\nabla u\|_{L^2}^2 . \tag{8.3}$$

Thus $\int_0^\infty \|\nabla u\|_{L^2}^2 \,\mathrm{d}t < \infty$.

8.2 The viscous lower bound

Theorem 8.1 (Small scales in the viscous Boussinesq system)**.** *Let $\nu > 0$ and $\Omega = \mathbb{T}^2$. Assume ρ_0 is even in x_1 and odd in x_2, u_{01} is odd–even, and u_{02} is even–odd. Suppose also that $\rho_0 \geq 0$ for $x_2 \geq 0$ and $\rho_0 = 0$ on the x_2-axis. Then the global smooth solution satisfies*

$$\limsup_{t\to\infty} t^{-1/6} \|\rho(t)\|_{\dot H^1(\mathbb{T}^2)} = \infty. \tag{8.4}$$

This is part of Kiselev et al. (2025). A second differentiation of E_P gives

$$E_P''(t) = A(t) - B(t) - \left\|\partial_{x_1}\rho(t)\right\|_{\dot H^{-1}}^2 , \tag{8.5}$$

where A is time-integrable by (8.3) and $\int_0^t B(s)\,\mathrm{d}s \lesssim t^{1/2}$ under a hypothetical uniform gradient bound. Therefore the negative Sobolev quantity must decay along a sequence. The Fourier-splitting estimate from the periodic IPM argument converts that decay into growth of $\|\rho\|_{\dot H^1}$, contradicting the hypothetical bound and yielding the algebraic exponent. A refined upper estimate remains subexponential, so the theorem separates genuine lower growth from known upper bounds.

8.3 The inviscid system and boundary-driven growth

For $\nu = 0$, taking curl gives

$$\partial_t\rho + u\cdot\nabla\rho = 0, \qquad \partial_t\omega + u\cdot\nabla\omega = -\partial_{x_1}\rho, \qquad u = \nabla^\perp(-\Delta)^{-1}\omega. \tag{8.6}$$

Global regularity on $\mathbb{R}^2$ or $\mathbb{T}^2$ remains open. On a strip, however, boundary and sign information yield unconditional lower bounds during the lifespan of a smooth solution.

Theorem 8.2 (Growth in a strip)**.** *Let $\Omega = \mathbb{T}\times[0,\pi]$. Assume ρ_0 is even in x_1, ω_0 is odd in x_1, and the initial vorticity integral over the right half-strip is nonnegative. Suppose there is $k_0 > 0$ such that $\rho_0 \geq k_0$ on $\{0\}\times[0,\pi]$ and $\rho_0 \leq 0$ on $\{\pi\}\times[0,\pi]$. During the smooth lifespan,*

$$\|\omega(t)\|_{L^p(\Omega)} \gtrsim t^{3-2/p}, \tag{8.7}$$

$$\|u(t)\|_{L^\infty(\Omega)} \gtrsim t, \tag{8.8}$$

$$\sup_{0\leq\tau\leq t} \|\nabla\rho(\tau)\|_{L^\infty(\Omega)} \gtrsim t^2. \tag{8.9}$$

Indeed, for the right half-strip Q,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_Q \omega\,\mathrm{d}x = \int_0^\pi \rho(0,x_2,t)\,\mathrm{d}x_2 - \int_0^\pi \rho(\pi,x_2,t)\,\mathrm{d}x_2 \geq k_0\pi. \tag{8.10}$$

Stokes' theorem converts this into a linearly growing circulation. Energy keeps $\|u\|_{L^2}$ bounded, so the circulation must concentrate in a thin region; interpolation then yields the stated vorticity and density-gradient bounds.

On $\mathbb{T}^2$, the same authors construct a broad class of symmetric sign-definite data such that

$$\sup_{0\le\tau\le t} \|\nabla\rho(\tau)\|_{L^\infty(\mathbb{T}^2)} \gtrsim t^{1/2} \tag{8.11}$$

during the lifespan. A parallel circulation argument applies to axisymmetric Euler flow in an annular cylinder

$$\Omega = \{(r,\theta,z) : r \in [\pi, 2\pi],\ \theta, z \in \mathbb{T}\}, \tag{8.12}$$

and gives $\left\|\omega^\theta(t)\right\|_{L^p} \gtrsim t^{3-2/p}$ together with $\|u(t)\|_{L^\infty} \gtrsim t$ under the corresponding symmetry and sign assumptions.

9 Synthesis and open directions

The examples above share a compact proof architecture:

1. identify a conserved or monotone weak quantity;
2. prove coercivity near a coherent structure, or obtain an integrable negative norm;
3. locate a geometric obstruction—a saddle point, a persistent level set, radial escape, or boundary circulation;
4. convert that obstruction into growth of a stronger norm.

For Euler dipoles, energy and rearrangement provide confinement while a saddle point creates gradients. For free-boundary Euler, total energy controls the geometry well enough to recover an approximate Biot–Savart law. For axisymmetric Euler, a transported relative vorticity couples radial escape to stretching. For IPM and Boussinesq, potential energy or circulation forces small scales even when global regularity is unknown.

> **Question 9.1** (A common research program)**.** Which relative equilibria are orbitally stable in a frame with enough temporal regularity to preserve hyperbolic flux? Which monotone quantities detect mixing without requiring a boundary? Can lower growth exponents be matched to the best known upper bounds while retaining an open set of initial data?

The questions in this section and earlier boxes are those emphasized during the June 2026 lectures. They should not be read as a claim about developments after the lecture dates.

A Source-page concordance

The following table records how every source page is represented. Repeated slides are counted once in the notes but listed in both source ranges.

Source	Pages	Destination in these notes
Day 1	1–4	Overview and 2D Euler framework (equation (1.2))
Day 1	5–8	Radial stability, Lamb dipoles, and energy coercivity
Day 1	9–23	Odd–odd quadrupoles, kernel estimate, and global concentrated dynamics
Day 1	24–26	Multiple dipoles; repeated and completed by Day 2, pp. 2–5
Day 2	1–5	Multiple-dipole decomposition, bootstrap, and Lagrangian crossing argument
Day 2	6–17	Gradient-growth survey, moving frames, generic mechanism, and open questions
Day 3	1–13	Fixed-domain benchmark and free-boundary Euler growth
Day 3	14–27	Axisymmetric Euler, moments, norm growth, mass escape, and open questions
Day 4	1–20	IPM formulation, small scales, and stratified-state instability
Day 4	21–32	Viscous/inviscid Boussinesq and annular-cylinder analogue
Day 4	33	Closing slide; no mathematical content

References

Ken Abe and Kyudong Choi. Stability of Lamb dipoles, 2022.

Ken Abe, In-Jee Jeong, and Yao Yao. Stability for multiple Lamb dipoles, 2025.

Hajer Bahouri and Jean-Yves Chemin. équations de transport relatives à des champs de vecteurs non-lipschitziens et mécanique des fluides. *Archive for Rational Mechanics and Analysis*, 127(2):159–181, 1994. doi: 10.1007/BF00377659.

Stephen Childress. Growth of anti-parallel vorticity in Euler flows. *Physica D: Nonlinear Phenomena*, 237(14–17):1921–1925, 2008. doi: 10.1016/j.physd.2008.02.028.

Kyudong Choi, In-Jee Jeong, and Yao Yao. Stability of vortex quadrupoles with odd-odd symmetry, 2024.

Diego Córdoba, Francisco Gancedo, and Rafael Orive. Analytical behavior of two-dimensional incompressible flow in porous media. *Journal of Mathematical Physics*, 48(6), 2007. doi: 10.1063/1.2404593. Article 065206.

Juan Dávila, Manuel del Pino, Mónica Musso, and Shrish Parmeshwar. Global in time vortex configurations for the 2d Euler equations, 2023.

Sergey A. Denisov. Infinite superlinear growth of the gradient for the two-dimensional Euler equation. *Discrete and Continuous Dynamical Systems*, 23(3):755–764, 2009. doi: 10.3934/dcds.2009.23.755.

Khakim Egamberganov and Yao Yao. Growth estimates for axisymmetric Euler equations without swirl, 2025.

Stephen Gustafson, Evan Miller, and Tai-Peng Tsai. Growth rates for anti-parallel vortex tube Euler flows in three and higher dimensions. *Journal of Mathematical Fluid Mechanics*, 28(1), 2026. doi: 10.1007/s00021-026-01001-0. Article 19.

Thomas Y. Hou and Congming Li. Global well-posedness of the viscous Boussinesq equations. *Discrete and Continuous Dynamical Systems*, 12(1):1–12, 2005. doi: 10.3934/dcds.2005.12.1.

Zhongtian Hu, Chenyun Luo, and Yao Yao. Small scale creation for 2d free boundary Euler equations with surface tension. *Annals of PDE*, 10(2), 2024. doi: 10.1007/s40818-024-00179-8. Article 21.

In-Jee Jeong, Yao Yao, and Tao Zhou. Superlinear gradient growth for 2d Euler equation without boundary, 2025.

Alexander Kiselev and Vladimir Šverák. Small scale creation for solutions of the incompressible two-dimensional Euler equation. *Annals of Mathematics*, 180(3):1205–1220, 2014. doi: 10.4007/annals.2014.180.3.9.

Alexander Kiselev and Yao Yao. Small scale formations in the incompressible porous media equation. *Archive for Rational Mechanics and Analysis*, 247(1), 2023. doi: 10.1007/s00205-022-01830-z. Article 1.

Alexander Kiselev, Jaemin Park, and Yao Yao. Small scale formation for the 2-dimensional Boussinesq equation. *Analysis & PDE*, 18(1):171–198, 2025. doi: 10.2140/apde.2025.18.171.

Deokwoo Lim and In-Jee Jeong. On the optimal rate of vortex stretching for axisymmetric Euler flows without swirl, 2024.

Y. H. Wan and M. Pulvirenti. Nonlinear stability of circular vortex patches. *Communications in Mathematical Physics*, 99(3):435–450, 1985. doi: 10.1007/BF01240356.

Xukai Yan and Yao Yao. Sharp stability for the interaction energy. *Archive for Rational Mechanics and Analysis*, 246(2–3):603–629, 2022. doi: 10.1007/s00205-022-01823-y. See also arXiv:2008.07502.

V. I. Yudovich. Non-stationary flows of an ideal incompressible fluid. *USSR Computational Mathematics and Mathematical Physics*, 3(6):1407–1456, 1963. doi: 10.1016/0041-5553(63)90247-7.

Andrej Zlatoš. Exponential growth of the vorticity gradient for the Euler equation on the torus. *Advances in Mathematics*, 268:396–403, 2015. doi: 10.1016/j.aim.2014.08.012. arXiv:1310.6128.

Andrej Zlatoš. Maximal double-exponential growth for the Euler equation on the half-plane, 2025.